

Incidence and Natural History of Idiopathic Chronic Inflammatory Demyelinating Polyneuropathy: A Population-Based Study in Iceland

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Key Words

Chronic inflammatory demyelinating polyneuropathy · Polyneuropathy · Epidemiology · Iceland · Survival

Abstract

Background and Aim: We report a population-based study conducted in Iceland to determine the incidence, clinical characteristics and prognosis of idiopathic chronic inflammatory demyelinating polyneuropathy (CIDP) during a 21-year period. **Methods:** Cases were identified from the records of all practicing neurologists in the country, the only neurology department in the country and both neurophysiology laboratories. All index cases met the 2010 European Federation of Neurological Societies/Peripheral Nerve Society criteria for CIDP. **Results:** Nineteen individuals fulfilled the diagnostic criteria during the study period. The average annual incidence was 0.3/100,000 (95% CI 0.04–2.47). There were 14 men (74%) in a gender ratio of 1:2.8. The mean age at diagnosis was 57 (range 19–81 years): women, 36 years and men, 63 years; $p = 0.0006$. The disease course was remitting-relapsing in 21% and chronic progressive or monophasic in 79%. The average length of follow-up was 6.9 years. The standardized mortality ratio for the 21-year study period was 0.9 (95% CI 0.3–2.2). **Conclusion:** We believe we have identified all diagnosed with CIDP in Iceland during a 21-year pe-

riod. Many had no or only limited disease progression over the years and mortality is not increased compared with the general population.

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Introduction

Chronic inflammatory demyelinating polyneuropathy (CIDP) is an acquired inflammatory peripheral neuropathy, usually presenting with a variable form of severe symmetric distal limb weakness with relative preservation of muscle bulk and impaired proprioception with relative preservation of pain and temperature sensation. The symptoms typically progress over weeks or months and the course can be remitting-relapsing (RR) or non-RR [1].

Recent population-based studies report an incidence of 0.4–1.6 per 100,000 population, but inclusion criteria vary [2–4]. Few studies have reported long-term outcome of CIDP patients [2, 5, 6].

We conducted a population-based study of CIDP in Iceland during a 21-year period to determine the incidence and the disease course, response to treatment and survival of the incidence cohort, in an island population.

Table 1. Clinical grading scale, adopted from Hahn et al. [9]

Clinical grading score	Symptoms
0	No symptoms
1	No disability, minor sensory signs or areflexia
2	Mild disability, ambulatory for >200 m without stick, mild weakness in one or more limb
3	Moderate disability, ambulatory for >50 m without stick, MRC grade 4
4	Severe disability, able to walk >10 m with support of stick
5	Requires support to walk 5 m
6	Cannot walk 5 m, able to stand unsupported and transfer to wheelchair, able to feed independently
7	Bedridden and severe quadriparesis, maximum MRC grade 3
8	Respirator dependent and quadriparesis, maximum MRC grade 2
9	Respirator dependent and quadriplegia
10	Dead

MRC = Medical research council.

Material and Methods

The study period was 21 years from January 1, 1991 through December 31, 2011. The average population of Iceland during the study period was 284.082 and there were around 15 practicing neurologists in Iceland, one neurological department and 2 neurophysiological laboratories [7].

We made an effort to identify all residents of Iceland diagnosed with CIDP during the study period. Case finding was done using multiple approaches including records of (a) Landspítali University Hospital, the only neurology department in Iceland; (b) all practicing neurologists; and (c) all neurophysiology laboratories.

Included in the study were all those who fulfilled the 2010 European Federation of Neurological Societies/Peripheral Nerve Society (EFNS/PNS) diagnostic criteria for definite or probable CIDP [8]. Those with possible CIDP were not included to increase the specificity of the diagnosis. We excluded all those who had a history or had clinical features suggestive of hereditary disease or other potential causes for polyneuropathy, including medications, toxins and infections.

All patient records were reviewed to verify diagnosis, determine gender and age at diagnosis, type of symptoms, time of diagnosis (when first seen by a neurologist for the symptoms), results of diagnostic studies, disease course and response to therapy. The course was considered RR if at least 2 episodes of rapid clinical worsening were documented and non-RR if symptoms worsened steadily during the follow-up period, remained stable or improved after the initial phase, without further relapses. Disability was assessed using a clinical grading scale (table 1) [9].

Treatment response and disease activity at the end of the follow-up period were determined using the modified CIDP dis-

ease activity status (CDAS) [5]. The study was approved by Icelandic National Bioethics Committee and Data Protection Authority.

Epidemiologic and Statistical Analysis

The average annual incidence was determined using the mid-year population for each year during the study period [7]. Confidence intervals were calculated assuming the Poisson distribution. Age and sex adjustment was done using the direct method. Simple linear regression analysis was used to compare incidence rates between 4 equally long blocks of time during the study period to examine possible bias due to for example, increasing awareness of the disease [3].

The Student's *t* test for independent samples was used to compare means. Pearson product moment correlation was used to determine correlation between 2 sets of data. Standardized mortality ratio (SMR) was determined using life tables. Calculations were done in Microsoft Excel 2011 and SPSS 20.

Results

Cases

We identified 19 residents of Iceland who fulfilled the 2010 EFNS/PNS diagnostic criteria for definite (*n* = 17) or probable (*n* = 2) CIDP during the study period. We did not identify any associated diseases with a known predisposition to demyelinating polyneuropathy.

Men were 74% (14/19) for a gender ratio of 1:2.8. All were registered residents of Iceland at the time of diagno-

sis, but one of them was Portuguese and had lived in Iceland for a few years at the time of diagnosis. The mean age at diagnosis was 57 (range 19–81 years); women, 36 years and men, 63 years; $p = 0.0006$.

Diagnostic Studies

Electrophysiological studies were available for all. Findings consistent with demyelination were present in all (table 2) and 84% (16/19) did not have evidence for acute denervation.

Cerebrospinal fluid (CSF) analysis was done at diagnosis in 63% (12/19), and 75% (9/12) of them had elevated protein (>450 mg/l). The CSF analysis was done after several months to years in an additional 26% of patients (5/19), and 80% (4/5) of them had elevated protein. Two did not have a CSF analysis done. Two patients had sural nerve biopsy done, both showing a normal nerve.

Incidence

The average annual incidence was 0.3 per 100,000 person years of observation (95% CI 0.04–2.47). There was no evidence for time trend for incidence in the study.

Clinical Characteristics

The patients were divided up into 4 clinical forms as follows: (i) coexisting proximal and distal weakness and distal sensory loss, 53% (10/19); (ii) predominantly distal weakness, 37% (7/19); (iii) predominantly sensory symptoms (1), and (iv) motor symptoms only (1). Impaired position sense was a prominent feature in 57% (11/19). Phrenic nerve paralysis was seen in 2.

Disability

At diagnosis, 74% (14/19) were ambulatory without aid, in spite of mild to moderate disability. Walking aid was needed by 16% (3/19) and another 10% (2/19) were confined to the wheelchair or were bedridden. The progression of disability during follow-up is shown in table 3.

Disease Category

The disease was RR in 21% (4/19) and non-RR in 79% (15/19). The non-RR group had a chronic progressive course in 53% (8/15) and monophasic course (subacute or acute worsening followed by a stable period) in 47% (7/15). The mean age at diagnosis for patients with RR disease was 50 (range 41–68; SD 12) and 59 (range 19–81; SD 20) for patients with non-RR course ($p = 0.4$). Patients with non-RR disease had a lower disability score at diagnosis, although not statistically significant (2.8 vs. 3.5; $p = 0.5$).

Table 2. Electrophysiological results

Electrophysiological abnormalities present in at least one nerve	Number of patients, %
Prolonged distal latency ($\geq 50\%$ of normal)	11/19 (58)
Conduction velocity slowing ($\leq 70\%$ of normal)	17/19 (89)
F-waves ($n = 14$)*	10/14 (71)
Abnormally prolonged ($\geq 130\%$ of normal)	5/14 (36)
Absent	9/14 (64)
Partial conduction block	18/19 (95)
Definite ($\geq 50\%$ drop in amplitude)	13/19 (68)
Probable ($\geq 30\%$ drop in amplitude)	5/19 (26)
Temporal dispersion*	6/12 (50)

Electrophysiological abnormalities present in at least one nerve.

* Information on F-waves and temporal dispersion were not available for all.

Treatment and Response

Immunomodulating treatment was used in 90% (17/19) at some point during the first 8 years after presentation (table 3). This included 3 who were treated only at the time of diagnosis, while others required repeated treatments.

Treatment regimens included intravenous immunoglobulins (16/17), either alone (10/17) or in combination, high-dose corticosteroids (6/17) and plasmapheresis (4/17). A single patient each received one of the following: azathioprine, etanercept and cyclophosphamide.

Follow-Up and Prognosis

The average length of follow-up was 6.9 years (range 0.3–15; SD 3.5). The mean clinical grade for non-RR CIDP was 2.8 (15/15) at diagnosis, 1.9 (15/15) after 1 year; 2.0 (14/15) at 4 years; 2.3 (8/15) at 8 years. A similar pattern with the most severe affection initially was seen for the RR, but the numbers are smaller. At the end of follow-up, 4 patients (21%) were cured according to CDAS (stable ≥ 5 years off treatment), 3 (16%) were in remission (stable ≤ 5 years off treatment), 6 (32%) had active but stable disease (stable on treatment >1 year), 1 patient was showing improvement on treatment <1 year, 5 (26%) had unstable active disease (worsening on treatment). Neither age nor length of follow-up correlated with the disease activity ($r = 0.3$ and -0.03 , respectively).

Mortality

Five of the 19 patients died during the follow-up period. Two died from respiratory failure. One of them (patient 6 in table 3) had a chronic progressive non-RR CIDP

Table 3. Clinical course and treatment

Patient No.	Gender	Age at DX	Clinical course	Clinical grade*				Time and duration of treatment	Total length of follow-up, YRs
				at diagnosis	after 1 year	after 4 years	after 8 years		
1	M	76	Non-RR	2	2	2	3	IVIG (30 g × 3 every 3 MOs) first 5 YRs after DX	8
2	M	69	Non-RR	2	1	2	–	IVIG (30 g × 3 every 6–8 WKs) first 6 YRs after DX	7
3	M	40	Non-RR	2	2	2	2	IVIG (90 g × 2 every 6 WKs) first 2 YRs from DX. OCS first 8 YRs from DX. PP × 7 during 2 MOs at 2 years from DX. ETC 25 mg twice weekly from YR 5 to 15	15
4	M	57	Non-RR	2	2	2	–	IVIG (35 g × 3 every 6–8 WKs) first 6 YRs after DX	6
5	F	19	Non-RR	2	1	1	1	IVIG (25 g × 4) once at DX. No treatment after that	11
6	F	48	Non-RR	4	2	3	4	OCS for 8 YRs from DX. PP × 40 first YR after DX. IVIG (30 g × 3 every 7 WKs) for 2 YRs (YRs 4–6). CTX (500 mg every 4 WKs) for 2 YRs (YRs 7–9)	9
7	F	46	RR	7	2	1	–	IVIG (36 g × 3–4 every 4 WKs) for 7 YRs after DX	7
8	M	57	Non-RR	2	1	1	1	IVIG (30 g × 3 every 6 WKs) for 8 YRs after DX	8
9	M	69	Non-RR	2	2	2	–	No treatment	4
10	M	45	RR	2	3	–	–	IVIG (40 g × 5 every 6–8 WKs) and OCS for 2 YRs after DX	2
11	M	68	RR	2	1	1	3	Only treated during relapses 1st relapse at DX: OCS (6 MOs) 2nd relapse at YR 8: IVIG (30 g × 3) stopped after allergic reaction, OCS, AZA for 1 YR, 3rd relapse at YR 14: OCS, AZA, PP × 10	14
12	M	81	Non-RR	2	1	1	1	PP × 5 at DX, no treatment after that	8
13	M	79	Non-RR	5	3	3	3	IVIG (30 g × 5) at DX and OCS for 1 MO. No treatment after that	10
14	M	51	Non-RR	5	2	2	–	IVIG (30 g × 3) at DX, no treatment after that	4
15**	F	41	RR	3	–	–	–	IVIG (30 g × 3) at DX and after 6 WKs	0.25
16	M	58	Non-RR	6	4	–	–	No treatment	2
17	F	24	Non-RR	2	1	1	–	IVIG (30 g × 3 every 6–8 WKs) for 1 YR after DX. No further treatment	7
18	M	75	Non-RR	2	2	3	–	IVIG (30 g × 3 every 8 WKs) for 1 year after DX. OCS 6 for 3 months after DX. No further treatment	6
19	M	77	Non-RR	2	3	3	4	IVIG (30 g × 3) once 6 YRs after DX. No further treatment	9

* The clinical grade is not assessed during relapses.

** Died from suicide 3 months after diagnosis.

Shaded area: follow-up not available. M = Male; F = female; DX = diagnosis; OCS = corticosteroids; PP = plasmapheresis; ETC = etanercept; AZA = azathioprine; CTX = cyclophosphamide; MO = month; WK = week; YR = year; IVIG = intravenous immunoglobulins.

and died 8 years after diagnosis, never responding to treatment. The second patient (patient 11 in table 3) had RR CIDP and died during his third relapse 14 years after diagnosis. He had responded to treatment during the first 2 relapses but not during the last one. The other 3 patients died from conditions unrelated to CIDP. There were 5.3 expected deaths among the 19 cases during the observation period following diagnosis (SMR 0.9; 95% CI 0.3–2.2).

Discussion

This is a population-based study in a well-defined island population, which is a powerful method to eliminate selection bias giving an accurate picture of the spectrum of symptoms and findings of CIDP [10]. We believe we have ascertained all CIDP cases diagnosed in Iceland during the study period and fulfilling the inclusion criteria. All the index cases were examined and diagnosed by a neurologist and all had at least one neurophysiological study done. Most index cases were identified from more than one source.

We excluded the category of possible CIDP to avoid false positive cases and consequent decrease in specificity, as its diagnosis only requires that a single nerve is found affected on a nerve conduction study [8]. The electrodiagnostic criteria used have the reported sensitivity of 81.3% and specificity of 96.2% for definite and probable CIDP. The specificity for possible CIDP is only 69.2% [11].

An epidemiological study of CIDP from England (Leicestershire and Rutland counties) found an incidence of 0.7/100,000 using the 2005 EFNS/PNS criteria. The results are similar to the current study, even if not directly comparable, since they also included ‘possible’ CIDP [4].

Men were almost 3 times as the number of women (74% men). Other studies have found similar results: Italy (68% men) and England (73% men) [2, 4]. The average age at diagnosis was 57 for all, 36 years for women and 63 years for men ($p = 0.0006$). A prevalence study from Australia describes a similar pattern of women being significantly younger [12].

We found a non-RR disease course (76%) to be more common than RR, which is similar or higher than other studies have found: Italy (74%), Olmsted county (56%) and England (48%) [2–4].

Two of our patients (10%) had phrenic nerve palsy, which has increasingly been reported in CIDP [13, 14].

We report a mean follow-up period of 6.9 years. An international study reported an average follow-up time of 6.4 years for 106 cases, with 11% cured, 20% in remission, 44% with stable active disease, 7% showing improvements and 18% with unstable active disease at the end of follow-up [5]. This compares with 21, 16, 32, 5 and 26%, respectively, in the current study. A recent study (2013) from Turkey reported a 60% cure-rate at 10 years follow-up after treating patients with a 5-year regimen of high-dose intravenous methylprednisolone. They also reported that 27% had an unstable active disease compared to 26% in current study [6]. A study from Italy reported an average follow-up time of 7.3 years. They used the modified Rankin Scale to score disability. At the prevalence day, they found that 68% had mild disability, 21% had moderate disability and 12% were severely disabled, compared to 45, 45 and 10%, respectively, at 8 years follow-up in the current study [2].

The disease course (table 3) is generally one of improvement. The symptoms were most severe initially and only 2 patients had a significant progression over the years of follow-up. This was seen in both non-RR and RR CIDP. Thus, most of them only required brief or no treatment but around a quarter of our patients (26%) did not respond to therapy, which is similar to the findings of previous studies [5, 6].

The mortality as revealed by this study is not different from that expected when compared with the general population. This is, to our knowledge, the first study to report the observed and expected survival of patients with CIDP in a well-defined population. The main usefulness of this study is that it is population based and gives an accurate picture of the spectrum of clinical involvement in CIDP. There are very few population-based studies on CIDP in the literature, and most are of the same size as this study. The number of patients is, however, small and this limits the analysis of subgroups.

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Disclosure Statement

There were no competing interests.

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